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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Materials and Methods Patients and single-cell isolation This study was approved by the Ethics Committee of the Shanghai Cancer Center, Fudan University, China (No. 1711178-9).
Consent: Written informed consent was obtained from all patients.
Inclusion and Exclusion Criteria
Likely doublets expressing both epithelial and TBmyeloid cell markers were further excluded from malignant cell analysis.
Attrition
Six nasopharynx biopsies were obtained by endoscopy from four treatment-naive patients with pNPC and two patients with rNPC.
Sex as a biological variable
Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes pathway enrichment analyses and gene set enrichment analysis (GSEA) were performed using the clusterProfiler package version 3.6.0. 15 To establish a protein-protein interaction (PPI) network of DEGs identified in malignant cells, STRING 16 was used to retrieve interacting genes, and the Cytoscape software was used to visualize the network.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
By randomly selecting 50 cells of each immune cell type, we generated a single-cell reference file and calculated the numbers of individual cell types from the bulk RNA-seq data using CIBERSORTx (http:CIBERSORTx.stanford.edu).
Blinding
not detected.
Power Analysis

not detected.	
Replication	
not required.	
Data Information	
Identifiers: Bulk RNA-seq data analysis NPC bulk RNA-seq raw data were downloaded from the public Gene Expression Omnibus database (accession number: GSE102349).	GSE102349

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
scRNA-seq library construction and data preprocessing The Cell Ranger software pipeline (version 3.1.0) provided by 10x Genomics was used to demultiplex cellular barcodes, map reads to the genome and transcriptome using the STAR aligner, and down-sample reads as required to generate normalized aggregate data across samples, producing a matrix of gene counts versus cells.	STAR		Suggestion: STAR (RRID:SCR_004463)(link)
Differences in pathway activities scored per cell were calculated using the LIMMA package (version 3.38.3).	LIMMA		Suggestion: LIMMA (RRID:SCR_010943)(link)
Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes pathway enrichment analyses and gene set enrichment analysis (GSEA) were performed using the clusterProfiler package version 3.6.0. 15 To establish a protein-protein interaction (PPI) network of DEGs identified in malignant cells, STRING 16 was used to retrieve interacting genes, and the Cytoscape software was used to visualize the network.	clusterProfiler		Suggestion: clusterProfiler (RRID:SCR_016884)(link)
	STRING		Suggestion: STRING (RRID:SCR_005223)(link)
	Cytoscape		Suggestion: Cytoscape (RRID:SCR_003032)(link)
The CopyKAT R package (version 1.0.5) 20 was also used to confirm the CNV results without identifying the reference cell type.	CopyKAT		Suggestion: CopyKAT (RRID:SCR_024512)(link)
Cell-cell interactions between different cell types were calculated using CellPhoneDB (version 2.0) 24 to identify significant ligand-receptor pairs within each sample.	CellPhoneDB		Suggestion: CellPhoneDB (RRID:SCR_017054)(link)
Bulk RNA-seq data analysis NPC bulk RNA-seq raw data were downloaded from the public Gene Expression Omnibus database (accession number: GSE102349).	Gene Expression Omnibus		Suggestion: Gene Expression Omnibus (GEO) (RRID:SCR_005012)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
For normally distributed data, P values were evaluated using a two-sided Students t-test.	t-test
The log-rank test and Kaplan-Meier analysis were performed to evaluate the prognostic value of each cell type.	log-rank test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq